Ex. No. 05	IMPLEMENTATION OF NAVIGATION AND ROUTING
Date of Exercise	
Git Hub Host Link	

Aim

To implement navigation between multiple Flutter screens using navigation and routing techniques.

Description

This experiment demonstrates how to navigate between multiple screens in a Flutter application. Three screens, namely Home, Profile, and Details, are created. `Navigator.push()` and `MaterialPageRoute` are used for screen navigation, while `Navigator.pop()` is used to return to the previous screen. Named routes are also configured using the `routes` property of `MaterialApp`.

Procedure:

1. A new Flutter project named `navigation_app` was created using the Flutter command.
2. Three Dart files named `home.dart`, `profile.dart`, and `details.dart` were created inside the `lib` folder.
3. The Home screen was created with an "Go to Profile" button.
4. `Navigator.push()` with `MaterialPageRoute` was used to navigate from the Home screen to the Profile screen.
5. The Profile screen was created with a "View Details" button.
6. `Navigator.push()` was again used to navigate from the Profile screen to the Details screen.
7. `Navigator.pop()` was implemented in the Details screen to return to the previous screen.
8. Named routes were defined inside `MaterialApp` using the `routes` property for Home, Profile, and Details screens.
9. The application was executed using Flutter and tested in the Chrome browser.
10. The navigation flow was verified as Home → Profile → Details, with back navigation to the previous screen.

Program

Main.dart

```
import 'package:flutter/material.dart';
import 'home.dart';
import 'profile.dart';
import 'details.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      initialRoute: '/',
      routes: {
        '/': (context) => const HomePage(),
        '/profile': (context) => const ProfilePage(),
        '/details': (context) => const DetailsPage(),
      },
    );
  }
}
```

home.dart

```
import 'package:flutter/material.dart';

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Home'),
      ),
      body: Center(
        child: ElevatedButton(
```

```
          onPressed: () {
            Navigator.pushNamed(context, '/profile');
          },
          child: const Text('Go to Profile'),
        ),
      ),
    );
  }
}
```

profile.dart

```
import 'package:flutter/material.dart';

class ProfilePage extends StatelessWidget {
  const ProfilePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Profile'),
      ),
      body: Center(
        child: ElevatedButton(
          onPressed: () {
            Navigator.pushNamed(context, '/details');
          },
          child: const Text('View Details'),
        ),
      ),
    );
  }
}
```

details.dart

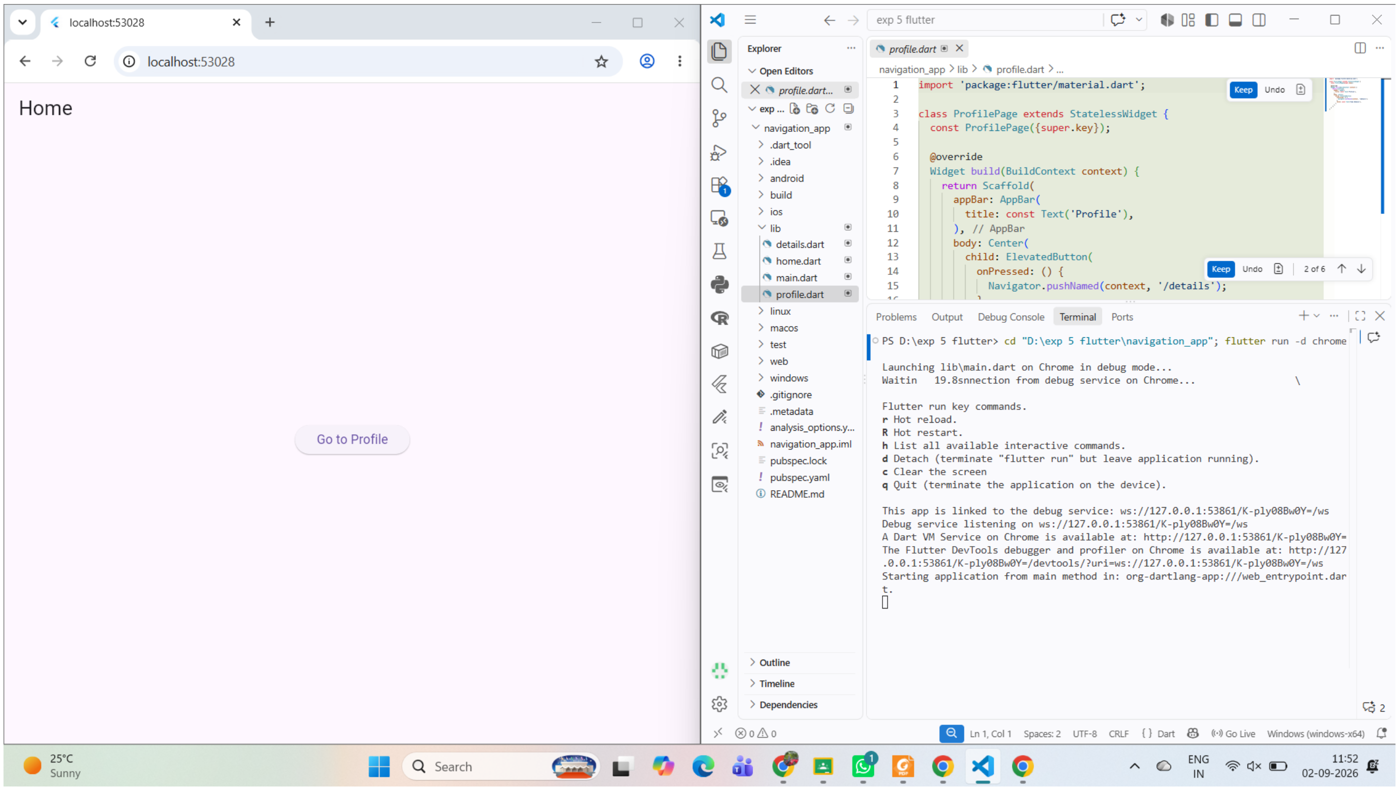
```
import 'package:flutter/material.dart';

class DetailsPage extends StatelessWidget {
  const DetailsPage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Details'),
      ),
      body: Center(
        child: ElevatedButton(
          onPressed: () {
            Navigator.pop(context);
          },
          child: const Text('Back to Profile'),
        ),
      ),
    );
  }
}

}
```

Output



Result

Thus, the Flutter application was successfully developed to implement navigation and routing between multiple screens. The application successfully navigated from Home → Profile → Details and allowed the user to return to the previous screen using back navigation.